

Flexibility

Properly managing a modern business is about making choices to create value. Valuation provides important insights for executives faced with making decisions on corporate strategy, acquisitions and divestments, capital structure, and other management actions. All these decisions take place against a backdrop of uncertainty about the outcomes of alternative courses of action.¹ However, in some cases, you can face decisions where not only is uncertainty present, but so is flexibility.

Managerial flexibility and uncertainty are not the same. In cases of uncertainty, the future of a company or a project may be extremely difficult to predict and depends on a single management decision—for example, to launch a new product line or to invest in a new production facility. Flexibility, in contrast, refers to choices managers may make between alternative plans in response to events. This is especially true when you are conducting valuations of investment projects.

The difference is important in deciding your approach to valuation. Whatever the degree of uncertainty, it is possible to value the asset in question by using a *standard* discounted-cash-flow (DCF) approach combined with either different scenarios or a stochastic simulation (see, for example, Chapter 17). But suppose management has planned to stage its investments in a business start-up. In that case, the managers may decide at each stage whether to proceed, depending on information arising from the previous stage. Where managers expect to respond flexibly to events, they need so-called *contingent valuation* approaches. These forecast, implicitly or explicitly, the future free cash flows, depending on the future states of the world and management decisions, and then discount these to today's value. For such decisions, alternative approaches provide more accurate valuation results and, perhaps even more important, deeper insights into what creates value.

Flexibility comes in many forms and can substantially alter the value of a business or project. But the business or project can have value only if

¹ See Chapters 4 and 13 for ideas on handling uncertainty, for example, with scenario-based approaches.

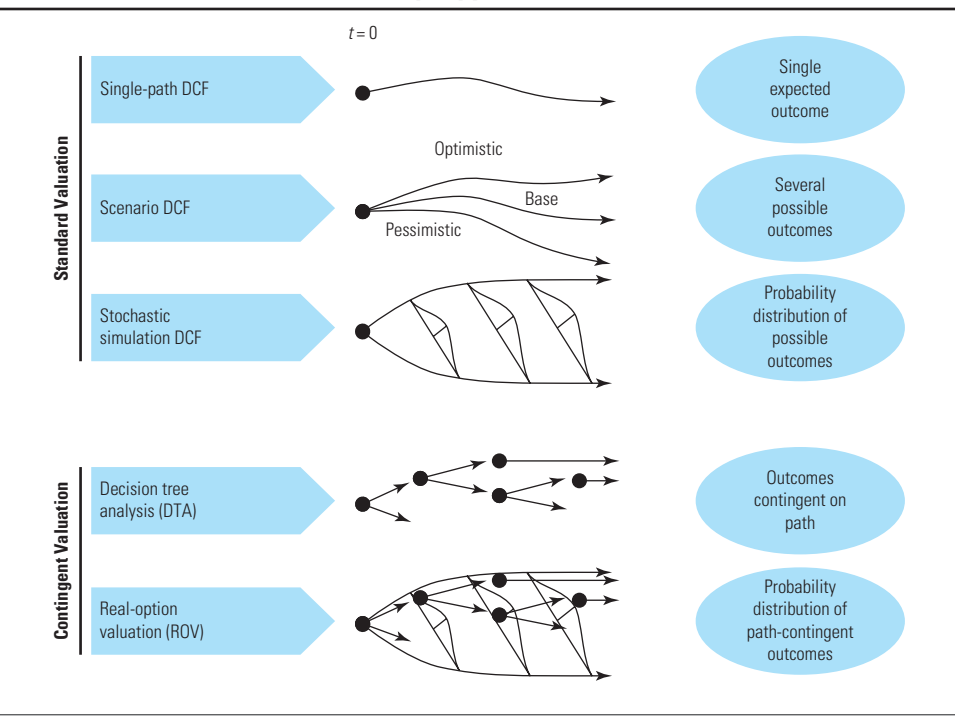
executives actively manage it to make better decisions. This chapter concentrates on the basic concepts of valuing managerial flexibility and real options in businesses and projects. It focuses on the following topics:

- Fundamental concepts behind uncertainty, flexibility, and value (when and why flexibility has value)
- Managing flexibility in terms of real options to defer investments; making follow-on investments; and expanding, changing, or abandoning production
- Comparison of decision tree analysis (DTA) and real-option valuation (ROV) to valuing flexibility, including situations in which each approach is more appropriate
- A four-step approach to analyzing and valuing real options, illustrated with numerical examples using ROV and DTA

A HIERARCHY OF APPROACHES

It is possible to illustrate a hierarchy of standard and contingent approaches to valuation under situations of uncertainty and flexibility and suggest when it is best to apply each (Exhibit 39.1). When a flexible response is neither expected

EXHIBIT 39.1 Valuation under Uncertainty: Approaches



nor required, you can choose from the following three variations of a standard DCF approach, depending on the level of uncertainty:

1. *Single-path DCF valuation.* When little uncertainty exists about future outcomes or when uncertainty is evenly spread around the expected outcomes, use a standard, single-path DCF analysis based on point estimates of future cash flows.
2. *Scenario-based DCF.* When significant uncertainty exists, especially when there is a possibility of much more upside than downside (or vice versa) in future cash flows, it is best to model future outcomes in two or more scenarios that capture the variation in the paths of future cash flow. This approach is easy to apply in, for example, valuing corporate or business strategies.
3. *Stochastic simulation DCF.* If you have reliable estimates about the underlying probability distributions of cash flows into the future, such as mean, standard deviation, and possibly skewness, it may be worthwhile to use a stochastic simulation DCF approach. In this approach, future cash flow paths are explicitly modeled and valued in a stochastic simulation. Because this approach is complex and requires voluminous data, applications are mostly restricted to specific industries, such as the valuation of insurance companies, and commodity-based businesses.

When managerial flexibility is called for, you need one of the following contingent valuation approaches, selected according to the amount of information available:

- *Decision tree analysis (DTA).* If there is limited information about the distribution of future cash flow paths and the decisions that management can take depending on these cash flows, use a decision tree analysis. As the following sections discuss, it builds on scenario DCF valuation and is straightforward and transparent. DTA is especially effective for valuing flexibility related to technological risks that are not priced in the market, such as investments in research and development (R&D) projects, product launches, and plant-decommissioning decisions.
- *Real-option valuation (ROV).* If you have reliable information about the underlying probability distributions of future cash flow paths, like those required for stochastic simulation, ROV could provide better results and insights. However, it requires sophisticated, formal option-pricing models that are harder for managers to decipher than DTA. The ROV approach is best suited to decisions in commodity-based businesses, such as investments in oil and gas fields, refining facilities, chemical plants, and power generators, because the underlying commodity risk is priced in the market.²

² See, for example, E. S. Schwartz and L. Trigeorgis, eds., *Real Options and Investment under Uncertainty: Classical Readings and Recent Contributions* (Cambridge, MA: MIT Press, 2001); T. Copeland and V. Antikarov, *Real Options: A Practitioner's Guide* (New York: Texere, 2003); or L. Trigeorgis, *Real Options: Managerial Flexibility and Strategy in Resource Allocation* (Cambridge, MA: MIT Press, 1996).

There are advantages to using either ROV or DTA, depending on the types of risks involved. In theory, ROV is more accurate. But it is not the right approach in every case. It cannot replace traditional discounted cash flow, because valuing an option using ROV still depends on knowing the value of the underlying assets. Unless the assets have an observable market price, you will have to estimate that value using traditional DCF.

Company-wide valuation models rarely take flexibility into account. To analyze and model flexibility accurately, you must be able to describe the set of specific decisions managers could make in response to future events and include the cash flow implications of those decisions. In valuing a company, flexibility therefore becomes relevant only in cases where management responds to specific events that may change the course of the whole company. For example, to value internet or biotech companies with a handful of promising new products in development, you could project sales, profit, and investments for the company as a whole that are conditional on the success of product development.³ Another example is a company that has built its strategy around buying up smaller players and integrating them into a bigger entity, capturing synergies along the way. The first acquisitions may not create value in their own right but may open opportunities for value creation through further acquisitions.

Flexibility is typically more relevant in the valuation of individual businesses and projects, as it mostly concerns detailed decisions related to production, capacity investment, marketing, research and development, and other factors.

UNCERTAINTY, FLEXIBILITY, AND VALUE

To appreciate the value of flexibility and its key value drivers, consider a simple example.⁴ Suppose you are deciding whether to invest \$6,000 one year from now to produce and distribute a new pharmaceutical drug already under development. In the upcoming final development stage, the product will undergo clinical tests on patients for one year, for which all investments have already been made. These tests involve no future cash flows. The trials could have one of two possible outcomes. If the drug proves to be highly effective, it will generate an annual net cash inflow of \$500 into perpetuity. If it is only somewhat effective, the annual net cash inflow will be \$100 into perpetuity. These outcomes are equally probable.

Based on this information, the expected future net cash flow is \$300, the probability-weighted average of the risky outcomes (\$500 and \$100). To keep it simple, we assume that success in developing the new product and the value

³ See, for example, E. S. Schwartz and M. Moon, "Rational Pricing of Internet Companies," *Financial Analysts Journal* 56, no. 3 (2000): 62–75; and D. Kellogg and J. Charnes, "Real-Options Valuation for a Biotechnology Company," *Financial Analysts Journal* 56, no. 3 (2000): 76–84.

⁴ The example is inspired by A. Dixit and R. Pindyck, *Investment under Uncertainty* (Princeton, NJ: Princeton University Press, 1994), 26.

of the new product are unrelated to what happens in the overall economy, so this risk is fully diversifiable by the company's investors. Therefore, the beta for this product is zero, and the cost of capital equals the risk-free rate—say, 5 percent. Assuming that the company will realize its first year's product sales immediately upon completing the trials and at the end of each year thereafter, the net present value (NPV) of the investment is estimated as follows:

$$\text{NPV} = \frac{-\$6,000}{1.05} + \sum_{t=1}^{\infty} \frac{\$300}{(1.05)^t} = \$286$$

To apply the NPV approach, we discount the incremental expected project cash flows at the cost of capital. Any prior development expenses are irrelevant, because they are sunk costs. Alternatively, if the project is canceled, the NPV equals \$0. Therefore, management should approve the incremental investment of \$6,000.

In this example of the NPV decision rule, undertaking development creates value. But there are more alternatives than deciding *today* whether to invest. Using an approach like the scenario approach described in Chapter 17, we can rewrite the previous NPV calculation in terms of the probability-weighted values of the drug, discounted to today:

$$\begin{aligned} \text{NPV} &= 0.5 \left[\frac{-\$6,000}{1.05} + \sum_{t=1}^{\infty} \frac{\$500}{(1.05)^t} \right] + 0.5 \left[\frac{-\$6,000}{1.05} + \sum_{t=1}^{\infty} \frac{\$100}{(1.05)^t} \right] \\ &= 0.5(\$4,286) + 0.5(-\$3,714) \\ &= \$286 \end{aligned}$$

Here, the NPV is shown as the weighted average of two distinct results: a positive NPV of \$4,286 following a favorable trial outcome and a negative NPV of -\$3,714 for an unfavorable outcome. If the decision to invest can be deferred until trial results are known, the project becomes much more attractive. Specifically, if the drug proves to be less effective, the project can be halted, avoiding the negative NPV. You invest only if the drug is highly effective, and the annual cash flow of \$500 more than compensates for the investment required. In practice, there would likely be an upfront investment need for the trial, regardless of its outcome, but we have abstracted from such costs to keep the example simple.

This flexibility is an option to defer the investment decision. To value the option, a contingent NPV approach can be used, working from right to left in the payoff tree shown in Exhibit 39.2.

$$\begin{aligned} \text{NPV} &= 0.5 \times \text{Max} \left[\left(\frac{-\$6,000}{1.05} + \sum_{t=1}^{\infty} \frac{\$500}{(1.05)^t} \right), 0 \right] \\ &\quad + 0.5 \times \text{Max} \left[\left(\frac{-\$6,000}{1.05} + \sum_{t=1}^{\infty} \frac{\$100}{(1.05)^t} \right), 0 \right] \\ &= 0.5(\$4,286) + 0.5(0) = \$2,143 \end{aligned}$$

EXHIBIT 39.2 Value of Flexibility to Defer Investment

\$				$t = 1$	$t = 2$	...	?
$t = 0$							
Contingent NPV = 2,143	$p =$	Successful product	Cash flow	500	500	...	500
		50%	Investment	(6,000)	—	...	—
	$1 - p =$	50%	Cash flow	100	100	...	100
		Unsuccessful product	Investment	(6,000)	—	...	—
Cost of capital = 5%							

Note: t = time, in years
 p = probability

The contingent NPV of \$2,143 is considerably higher than the \$286 NPV of committing today. Therefore, the best alternative is to defer a decision until the trial outcomes are known. The value of the option to defer investment is the difference between the value of the project with flexibility and its value without flexibility: \$2,143 – \$286 = \$1,857.

Based on this example, it is possible to summarize the distinction between the standard and contingent NPVs. The standard NPV is the maximum, decided today, of the expected discounted cash flows or zero:

$$\text{Standard NPV} = \text{Max}_{t=0} \left(\frac{\text{Expected (Cash Flows)}}{\text{Cost of Capital}}, 0 \right)$$

The contingent NPV is the expected value of the maximums, decided when information arrives, of the discounted cash flows in each future state or zero:

$$\begin{aligned} \text{Contingent NPV} &= \text{Expected}_{t=0} \\ &\times \left[\text{Max} \left(\frac{\text{Cash Flows Contingent on Information}}{\text{Cost of Capital}}, 0 \right) \right] \end{aligned}$$

These two NPV approaches use information quite differently. Standard NPV forces a decision based on today's expectation of future information, whereas contingent NPV permits the flexibility of making decisions after the information arrives. Unlike standard NPV, it captures the value of flexibility. A project's contingent NPV will always be greater than or equal to its standard NPV.

The value of flexibility is related to the degree of uncertainty and the room for managerial reaction (see Exhibit 39.3). It is greatest when uncertainty is high and managers can react to new information. In contrast, if there is little uncertainty, managers are unlikely to receive new information that would alter future decisions, so flexibility has little value. Similarly, if managers cannot act on new information that becomes available, the value of flexibility is low.

Including flexibility in a project valuation is most important when the project's standard NPV is close to zero—that is, when the decision whether to go ahead with the project is a close call. Sometimes senior management intuitively overrules standard NPV results and accepts an investment project for strategic reasons, for example, because the project creates an initial market position that can be expanded at a later stage if and when the company has

EXHIBIT 39.3 When Is Flexibility Valuable?

Ability to respond, room for managerial flexibility	High	Moderate flexibility value	High flexibility value
	Low	Low flexibility value	Moderate flexibility value
		Low	High
		Uncertainty, likelihood of receiving new information	

the competitive products or services to offer. In these cases, the flexibility recognized in contingent valuation fits better with strategic intuition than do the rigid assumptions of standard NPV approaches.

What Creates Flexibility Value

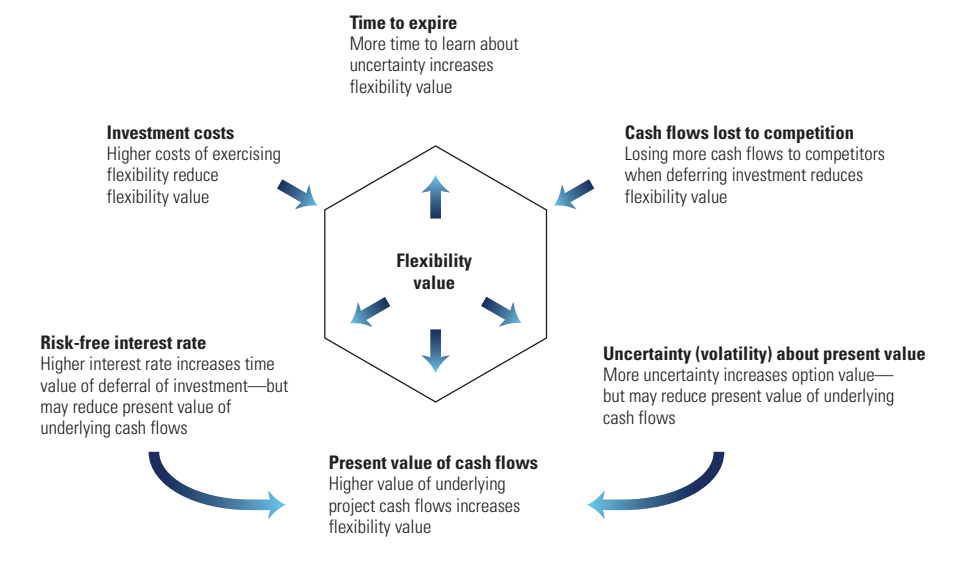
To identify and value flexibility, you must understand where its value comes from. Consider what happens if the range of possible annual cash flow outcomes (originally \$500 versus \$100 per year) increases to \$600 versus \$0. Since expected cash flows and cost of capital remain unchanged, the standard NPV is the same (\$286).⁵ However, the contingent NPV increases from its prior level of \$2,143:

$$\begin{aligned}
 \text{NPV} &= 0.5 \times \text{Max} \left[\left(\frac{-\$6,000}{1.05} + \sum_{t=1}^{\infty} \frac{\$600}{(1.05)^t} \right), 0 \right] \\
 &\quad + 0.5 \times \text{Max} \left[\left(\frac{-\$6,000}{1.05} + \sum_{t=1}^{\infty} \frac{\$0}{(1.05)^t} \right), 0 \right] \\
 &= 0.5(\$6,286) + 0.5(0) \\
 &= \$3,143
 \end{aligned}$$

The contingent NPV of \$3,143 is almost 50 percent greater at this higher level of uncertainty. Why? As in the original case, only the cash flows from the favorable outcome affect the contingent valuation. Since these cash flow projections have increased by 20 percent and the required investment has not changed, the contingent NPV increases substantially. The value of the deferral option rises from \$1,857 to \$2,857 (computed as \$3,143 – \$286).

⁵ We assume that the trial outcome risk is uncorrelated with the overall economy.

EXHIBIT 39.4 Drivers of Flexibility Value



We can formally derive the key value drivers of real options from the pricing theory of financial options such as, for example, call and put options on equity shares. In our original example, the deferral option is identical to a call option with an exercise price of \$6,000 and a one-year maturity on an underlying risky asset that has a current value of \$6,000 and a variance determined by the cash flow spread of \$400 across outcomes.⁶ As with financial options, the value of a real option depends on six drivers, summarized in Exhibit 39.4.

These drivers of option value show how allowing for flexibility affects the valuation of a particular investment project. Holding other drivers constant, option value decreases with higher investment costs and more cash flows lost while holding the option. Option value increases with higher value of the underlying asset's cash flows, greater uncertainty, higher risk-free interest rates, and a longer lifetime of the option. With higher option values, a standard DCF calculation that ignores flexibility will more seriously underestimate the true value of an investment project.

Be careful how you interpret the impact of value drivers when designing investment strategies to exploit flexibility. The impact of any individual driver described in Exhibit 39.4 holds only when all other value drivers remain constant. In practice, changes in uncertainty and interest rates not only affect the value of the option but usually change the value of the underlying

⁶ The current value of the underlying risky asset is the present value of expected annual cash flows of \$300 into perpetuity, discounted at a 5 percent cost of capital.

asset as well. When you assess the impact of these drivers, you should assess all their effects on the option's value, both direct and indirect. Take the case of higher uncertainty. In our example, we increased the uncertainty of future cash flows by widening the gap between future cash flows in the favorable and unfavorable scenarios from \$400 to \$600. But we kept the expected value of the future cash flows unchanged at \$300 so that their present value remained constant. However, if greater uncertainty lowers the expected level of cash flows or raises the cost of capital, the value of the underlying asset declines so that the impact on the value of the option could be negative. The same holds for the impact of an increase in the risk-free interest rate. Higher interest rates reduce the present value of the required investment, thereby increasing the option value—if the value of the underlying asset is assumed constant. But if higher interest rates lead to an increase in the cost of capital, the present value of cash flows on the underlying asset would decrease, which could lower the option's value.

MANAGING FLEXIBILITY

Contingent valuation is an important tool for managers trying to make the right decisions to maximize shareholder value when faced with strategic or operating flexibility. In actual practice, however, flexibility is never as well defined and straightforward as in the preceding examples. Much depends on management's ability to recognize, structure, and manage opportunities to create value from operating and strategic flexibility. A detailed discussion is beyond the scope of this book,⁷ but we provide some basic guidelines here.

To *recognize* opportunities for creating value from flexibility when assessing investment projects or strategies, managers should try to be as explicit as possible about the following details:

- *Events.* What are the key sources of uncertainty? Which events will bring new information and when? A source of uncertainty is important only if relevant new information about it is likely to trigger a decision change. For example, investing in a pilot project for a product launch makes sense only if there is a chance that the pilot outcome would actually change the launch decision. Similarly, options to switch inputs for manufacturing processes are valuable only if the input prices can be expected to diverge significantly.
- *Decisions.* What decisions can management make in response to events? It is important that managers have some discretion to react to a relevant event. For example, intense competition among smartphone

⁷ For a more in-depth discussion, see, for example, Copeland and Antikarov, *Real Options*, or Trigeorgis, *Real Options*.

manufacturers makes it unattractive for managers to defer a decision to launch new product versions with innovative features such as voice-control or foldable-screen technology until there is more information about potential demand for such features.

- *Payoffs*. What payoffs are linked to these decisions? Bear in mind that there should be a positive NPV to be captured in some realistic future state of the world. This NPV should be derived from sustainable competitive advantages. For example, some investors attribute high value to certain e-commerce start-ups as “options for future growth,” often based on multiples of enterprise value over unique website visitors per month. But website visits alone do not create value. Moreover, the value of e-commerce start-ups depends upon their future cash flows. Start-ups can represent valuable options only if they build sustainable, competitive business models in some plausible future scenarios. Valuing start-ups as options requires articulating what the scenarios are, as well as predicting their likelihood of success and associated businesses cash flows.

With regard to *structuring* flexibility, some projects or strategies have predefined, built-in flexibility. Take, for example, research and development (R&D) investments in pharmaceutical products where the outcomes of clinical or patient trials provide natural moments to decide whether to stop or proceed with investments. But in many other cases, flexibility can be incorporated into a project to create maximum value. One example would be redesigning infrastructure investments in ports or airfields in stages such that future expansion takes place only if and when needed. Another would be reshaping a growth strategy in such a way that it explicitly includes options to redirect resources as more information becomes available.

In the end, flexibility has value only if managers actually *manage* it—that is, use new information to make appropriate changes to their decisions. Therefore, companies should ensure that their managers face proper incentives to capture potential value from flexibility. For example, the option to pull out of a staged-investment project when intermediate results are disappointing has no value if managers do not act on the information. As is sometimes the case, managers will point to nothing more than large sunk costs as the rationale for their inaction. But they forget that value is determined only by future cash flows, so that sunk costs are always irrelevant. In the case where a company bases its strategy on creating growth options through a string of acquisitions, those options generate maximum value only if the company delays further acquisitions until new, positive information about their potential arrives. The company leaves the option value on the table if it proceeds with additional acquisitions in the dark.

EXHIBIT 39.5 **Classification of Real Options**

Option type	Financial equivalent	Definitions	Example(s)
Option to defer investment	Call option	The option to defer an investment until the present value of an asset rises above the development costs	The ability of a leaseholder of an undeveloped oil reserve to defer development and investment until oil prices have elevated the value of the reserves above their development costs
Abandonment option	Put option	The option to abandon a project if its present value falls below its liquidation value	Long-term rental leases of airplanes that give the lessee the flexibility to prematurely dissolve the contract and return the plane to the lessor at a prespecified termination fee
Follow-on (compound) option	Series of options on options	The option to invest in stages, contingent on performance	A factory, R&D program, new-product launch, or oil field built so that management can continue the project at each stage by investing additional funds (an exercise price) or abandon it for whatever they can fetch
Option to expand or contract	Call or put option	The option to resize an investment depending on performance	A production facility built so that it can be easily expanded or contracted if a product is more or less successful than anticipated
Option to extend or shorten	Call or put option	The option to shorten or extend the life of an asset or contract	Real estate leases with clauses that allow lessors to extend or shorten the term of the lease
Option to increase scope	Call option	The ability to increase or decrease the number of activities in the future	A hotel designed so that the owner can easily diversify beyond lodging services, such as by adding conference facilities
Switching options	Portfolio of call and put options	The ability to switch the operation of a project on and off—or to switch operations between two distinct locations	A flexible manufacturing system that can produce two or more different products, peak-load power generation, or the ability to exit and reenter an industry

To help managers recognize, structure, and manage opportunities for capturing value from flexibility, we segment options into the categories described in Exhibit 39.5 and provide some examples.

METHODS FOR VALUING FLEXIBILITY

As mentioned earlier in this chapter, the two methods for contingent valuation are decision tree analysis (DTA) and real-option valuation (ROV) using formal option-pricing models. We will illustrate each method with a simple example: the opportunity to invest \$105 at the end of one year in a mining project that has an equal chance of returning either \$150 or \$50 in cash flow, depending on the mineral price. The risk-free rate, r_f , is 5 percent, and the weighted average cost of capital (WACC) for the project is 10 percent. The present value (PV) of the cash flows today is:

$$PV = \frac{0.5(\$150) + 0.5(\$50)}{1.10} = \$90.9$$

EXHIBIT 39.6 **Contingent Payoffs for Investment Project, Twin Security, and Risk-Free Bond**

\$						
$t = 0$		$t = 1$				
			Project without flexibility	Project with flexibility	Twin security	Risk-free bond
NPV = ?	Successful project $p = 50\%$	Cash flow	150	150		
		Investment	(105)	(105)		
		Net cash flow	45	45	50	1.05
	Unsuccessful project $1 - p = 50\%$	Cash flow	50	50		
		Investment	(105)	(105)		
		Net cash flow	(55)	–	16.7	1.05

Risk-free rate = 5%
WACC = 10%

Note: t = time, in years
 p = probability

If an investment decision were required immediately, the project would be declined. The standard NPV of the mining project equals the discounted expected cash flow of \$90.90 minus the present value of the investment outlay of \$105 next year. Since the level of investment is certain, it should be discounted at the risk-free rate of 5 percent:

$$\text{Standard NPV} = \$90.9 - \frac{\$105}{1.05} = \$90.9 - \$100 = -\$9.1$$

The answer changes if management has flexibility to defer the investment decision for one year, allowing it to make the decision after observing next year's mineral price and the associated cash flow outcome (see Exhibit 39.6). The net cash flows in the favorable state are \$150 – \$105 = \$45. In the unfavorable state, management would decline to invest, accepting net cash flows of \$0.

To value this flexibility, we first use an ROV approach and then repeat the valuation with the DTA approach.

Real-Option Valuation

Option-pricing models use a *replicating portfolio* to value the project. The basic idea of a replicating portfolio is straightforward: if you can construct a portfolio of priced securities that has the same payouts as an option, the portfolio and option should have the same price. If the securities and the option are traded in an open market, this identity is required; otherwise arbitrage profits are possible. The interesting implication is that the ROV approach lets you correctly value complex, contingent cash flow patterns.

Returning to our \$105 investment project, assume there exists a perfectly correlated security (or commodity, in this example) that trades in the market for \$30.30

per share (or unit).⁸ Its payouts (\$50 and \$16.70) equal one-third of the payouts of the project, and its expected return equals the underlying project's cost of capital.

This twin security can be used to value the project, including the option to defer, by forming a replicating portfolio.⁹ Consider a portfolio consisting of N shares of the twin security and B risk-free bonds with a face value of \$1. In the favorable state, the twin security pays \$50 for each of the N shares, and each bond pays its face value plus interest, or $(1 + r_f)$. Together, these payouts must equal \$45. Applying a similar construction to the unfavorable state, we can write two equations with two unknowns:

$$\$50.0N + \$1.05B = \$45$$

$$\$16.7N + \$1.05B = 0$$

The solution is $N = 1.35$ and $B = -21.43$. Thus, to build a replicating portfolio, buy 1.35 shares and short 21.43 bonds (shorting a bond is common language for selling a bond, or borrowing money).

This position produces the same cash flow as the investment project under both states. Therefore, the value of the project, including the ability to defer, should equal the value of the replicating portfolio:

$$\begin{aligned}\text{Contingent NPV} &= N(\text{Price of Twin Security}) - B(\$1) \\ &= 1.35(\$30.3) - 21.43(\$1) \\ &= \$19.5\end{aligned}$$

The value of the deferral option is the difference between the total contingent NPV of the project and its standard NPV without flexibility: $\$19.50 - (-\$9.10) = \$28.60$ (remember, the standard NPV was negative).

Contingent NPV can also be determined with an alternative ROV approach called *risk-neutral valuation*. The name is somewhat misleading because a risk-neutral valuation does adjust for risk, but as part of the scenario probabilities rather than the discount rate. To value an option, weight the future cash flows by risk-adjusted (or so-called risk-neutral) probabilities instead of the actual scenario probabilities. Then discount the probability-weighted average cash flow by the risk-free rate to determine current value. The risk-neutral probability of the favorable state, p^* , is defined as follows:¹⁰

$$p^* = \frac{1 + r_f - d}{u - d} = 0.45$$

⁸ You could also use this twin security to value the investment project without flexibility by means of a replicating portfolio. Because the twin security's cash flows are always exactly one-third of the project cash flows, the project without flexibility should be worth three times as much as the twin security, or \$90.90 ($= 3 \times \30.30). The twin security is a basic concept that is implicitly used in standard DCF as well; you derive the beta of a project by identifying a highly correlated traded security and use that security's beta as input for the cost of capital in the DCF valuation.

⁹ If the project itself were traded, you would not need a twin security but would construct a replicating portfolio with the traded value of the project itself, as in the case of financial options on traded stocks.

¹⁰ See, for example, Trigeorgis, *Real Options*, 75–76.

where

$$u = \frac{\text{FV(Favorable State)}}{\text{PV}} = \frac{\$50.0}{\$30.3} = 1.65$$

$$d = \frac{\text{FV(Unfavorable State)}}{\text{PV}} = \frac{\$16.7}{\$30.3} = 0.55$$

Solve by substituting:

$$p^* = 0.45$$

$$1 - p^* = 0.55$$

These probabilities implicitly capture the risk premium for investments perfectly correlated with the twin security. We discount the future cash flows weighted by the risk-neutral probabilities at the risk-free rate of 5 percent, arriving at exactly the same value determined using the replicating portfolio:

$$\text{Contingent NPV} = \frac{0.45(\$45) + 0.55(0)}{1.05} = \$19.5$$

It is no coincidence that the replicating portfolio and risk-neutral valuation lead to the same result. They are mathematically equivalent, and both rely on the price of the twin security to derive the value of an investment project with an option to defer.

Valuation Based on Decision Tree Analysis

A second method for valuing a project with flexibility is to use DTA. This leads to the right answer in principle, but only if we apply the correct cost of capital for a project's contingent cash flows.

One DTA approach is to discount the project's contingent payoffs net of the investment requirements. Unfortunately, we can only derive the correct cost of capital for these cash flows from the ROV results. Given the project's contingent NPV of \$19.50 with equal chances of paying off \$45 or \$0, the implied discount rate from the ROV analysis is 15.5 percent.¹¹ This is significantly above the underlying asset's 10 percent cost of capital, because the contingent cash flows are riskier. The contingent NPV has an equal chance of increasing by 131 percent or decreasing by 100 percent. The value of the underlying asset (\$90.90) has a 50–50 chance of going up 65 percent (to \$150) or down 45 percent (to \$50). If the underlying asset's cost of capital of 10 percent were used, the DTA results would therefore be too high relative to the correct ROV result:

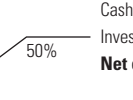
$$\text{Contingent NPV} = \frac{0.5(\$45) + 0.5(0)}{1.10} = \$20.5$$

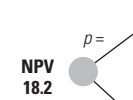
¹¹ In this simplified example, there is one value for the cost of capital. In general, the cost of capital for the contingent cash flows is not constant. It changes with the risk of the option across time and states of the world.

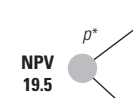
$$\text{Contingent NPV} = 0.5 \left(\frac{\$150}{1.10} - \frac{\$105}{1.05} \right) + 0.5(0) = \$18.2$$

For longer-term contingent payoffs, this DTA approach generates results that are closer to the correct ROV value. The next section discusses how this second DTA approach can lead to the exact ROV outcome if the underlying risk is either diversifiable or nondiversifiable but is too small to influence the future investment decision (that is, if the project value would exceed the investment requirements even in the unfavorable state).

As summarized in Exhibit 39.7, the standard NPV approach undervalues our mining project at $-\$9.10$. The ROV approach generates a correct value (NPV = $\$19.50$) because it captures the value of flexibility by using a replicating portfolio or risk-neutral valuation. The DTA approach at $\$18.20$ is quite close in this example, capturing almost the entire gap between the standard NPV valuation and the more granular ROV result. But the DTA results might

Standard NPV			
 NPV (9.1)	Cash flow	150	
	Investment	(105)	
	Net cash flow	45	
	Cash flow	50	
	Investment	(105)	
	Net cash flow	(55)	
Risk-free rate = 5% WACC = 10%			

Contingent NPV			
Decision tree analysis ¹			
 NPV 18.2	Cash flow	150	
	Investment	(105)	
	Net cash flow	45	
	Cash flow	50	
	Investment	(105)	
	Net cash flow	–	

Real-option valuation ²			
 NPV 19.5	Cash flow	150	
	Investment	(105)	
	Net cash flow	45	
	Cash flow	50	
	Investment	(105)	
	Net cash flow	–	

² Using risk-neutral valuation.

EXHIBIT 39.8 **Application Opportunities for Real-Option Valuation vs. Decision Tree Analysis**

Available data	Nontraded assets	Decision tree analysis	Decision tree analysis, real-option valuation
	Traded assets	Decision tree analysis	Real-option valuation
		Diversifiable	Nondiversifiable
		Underlying risk	

be further off or closer to the ROV mark, depending on the project’s payoffs and risks.

This example does not mean that ROV is always the best approach to valuing managerial flexibility. The stylized example did not take into account two important aspects of real-life investment decisions: the type of prevailing risk and the availability of data on the value and variance of cash flows from the underlying asset. Exhibit 39.8 identifies when each method is most suitable. As we explain next, the more straightforward DTA is often the better approach because in practice (most of) the underlying risk is diversifiable or because only rough estimates are available for required inputs such as the underlying asset value and variance. In addition, DTA is easier to use and understand. ROV works best only when the future cash flows are closely linked to traded commodities, securities, or currencies. Not surprisingly, real-option valuations are most often used for commodity-linked investments, such as in the mining and oil industries.

Prevailing Risk: Diversifiable and Nondiversifiable Investment projects can be exposed to a wide range of risks, such as product price and demand risk, interest and currency risks, technological risk, and political risk. The question is which particular risk (or group of risks) is *prevailing*—in other words, which risk could affect a project’s cash flow to such an extent that it would change management’s future decisions. The following examples of prevailing risks describe whether the risks are diversifiable and how this affects the choice of a valuation tool:

- If commodity prices (as in mining, the oil industry, or power generation) or currency and interest rates are keys to future investment decisions, the prevailing risk *is not diversifiable*, and only ROV leads to the theoretically correct valuation. This was illustrated in the previous example

in this chapter, where the difference in mining payoffs stemmed from changes in the mineral price. The DTA approach could not provide a correct value, although it was quite close in that particular case.

- If technological risks (such as customer preferences, technological innovations, drug trial outcomes, or geological survey results) are critical to future investment decisions, the prevailing risk is *diversifiable*, and both ROV and DTA are effective tools for valuing flexibility. In our experience, this is the more common case for prevailing risk. Applying DTA, it is possible to discount the project's payoffs in each scenario at the cost of capital of the underlying asset and discount the investment requirements at the risk-free rate (see Equation 39.1 in the example of the pharmaceutical drug company presented near the beginning of the chapter).¹²

Let's illustrate how you can apply ROV and DTA, depending on which group of risks dominates in a more complex version of the mining example from the previous section. In addition to price risk, there is now also uncertainty about the size of the reserves found (see Exhibit 39.9, Example 1). The mine is either very large, with reserves at 2.50 times the expected level, or very small, at 0.26 times the expected level. The probability of very large mining reserves is 33 percent, versus a probability of 67 percent for very small reserves. This implies that there are not just two but four possible outcomes to the initial investment, with cash flows ranging from \$375 in the large-mine and high-mineral-price scenario ($\$150 \times 2.50$) to \$13 in the small-mine and low-price scenario ($\$50 \times 0.26$). As the conditional payoffs in the exhibit reflect, the rational decision is to start production only if the mine turns out to be large, regardless of the commodity price. For a small mine, even the high-price scenario does not justify production, as the investment requirements (\$105) exceed the cash flow ($\$39 = \150×0.26).

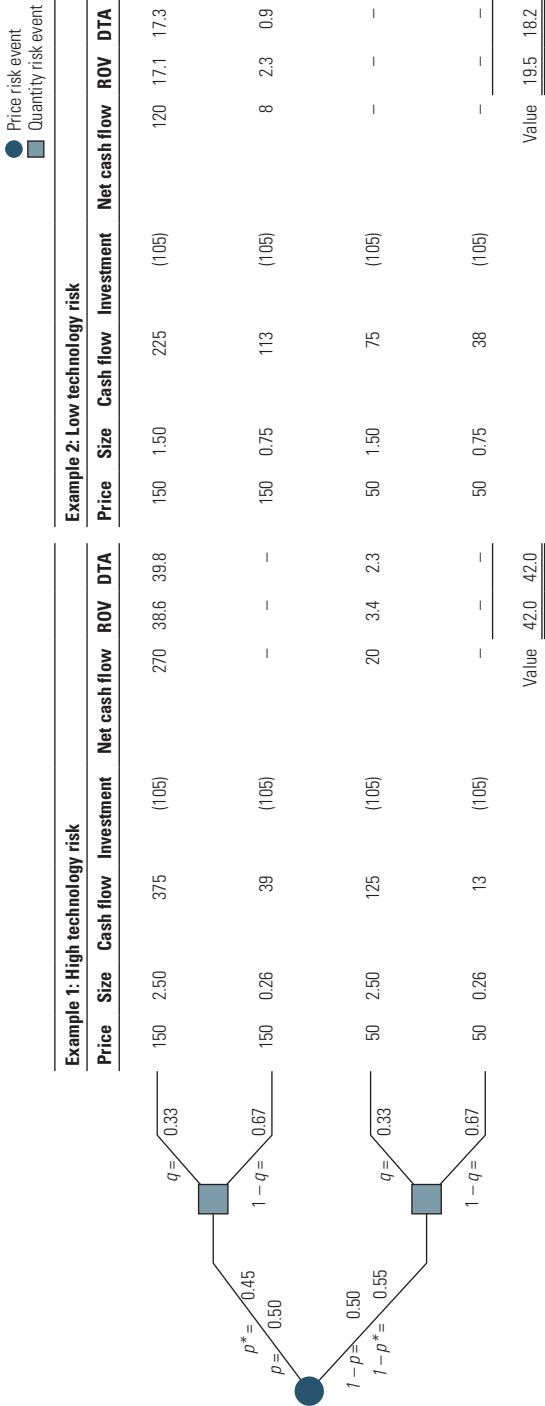
To use the ROV approach to derive the valuation results, multiply the conditional payoffs by the risk-neutral (pseudo-)probabilities for the price scenarios and the normal probabilities for the quantity scenarios, and then discount at the risk-free rate.¹³ For example, for the large-mine, high-price scenario:

$$\frac{0.45 \times 0.33 \times \$270}{1 + 5\%} = \$38.6$$

¹² To value the drug development project with an ROV approach, we build a replicating portfolio. Assume a twin security exists whose payoffs are perfectly correlated with the outcome of the drug trial, generating \$52.50 when the outcome is favorable and \$10.50 when it is unfavorable. Because its cash flows are driven by technological risk only, the security's market beta is zero, and its present value must be \$30. A replicating portfolio consists of a long position of 107.1 of these securities and a short position of \$1,071.40 in risk-free bonds. The ROV is therefore $107.1(\$30) - \$1,071.4(1) = \$2,143$. See also Dixit and Pindyck, *Investment under Uncertainty*, 30–32, for a similar proof.

¹³ We can use the risk-neutral probabilities from the original example because the mineral price risk has not changed. For the quantity risk, no risk adjustment to the probabilities is needed because it is diversifiable. We used the risk-neutral probability approach for the ROV valuation because it is more straightforward to apply here; of course, the replicating portfolio approach leads to an identical value.

EXHIBIT 39.9 Valuation Result for Mixed Price and Quantity Risk



Summing these values over all possible scenarios leads to an ROV of \$42.¹⁴

Taking a DTA approach instead, we would multiply the conditional payoffs by the normal probabilities for price and quantity scenarios and then separately discount the cash inflows at the mine's cost of capital and the investment cash flow at the risk-free rate. For the large-mine, high-price scenario, for example:

$$0.50 \times 0.33 \times \left(\frac{\$375}{1 + 10\%} - \frac{\$105}{1 + 5\%} \right) = \$39.8$$

Similarly, summing the results over all scenarios, we obtain a DTA value of \$42, which is exactly equal to the ROV result. The reason is that the decision is driven entirely by the diversifiable, technological risk related to the size of mine. The nondiversifiable price risk leads to different cash flows but does not matter for the investment decision.¹⁵

The DTA and ROV approaches both provide the theoretically correct answer when the contingent decisions are (predominantly) driven by diversifiable underlying risk. Examples are geological risks, such as the size of an undeveloped oil field, and even some forms of marketing risk, such as consumer acceptance of a new product. As in the numerical illustration, these risks often have more impact on value than nondiversifiable risks. For example, the driver of the decision to invest in drug development is whether the drug passes the trials, not whether the drug—once successfully developed—is worth more or less, depending on general economic conditions.

In contrast, only the ROV approach is theoretically correct if the contingent decision is affected by nondiversifiable risk; the DTA result would lead to an approximate result. For the same numerical illustration as before, the nondiversifiable price risk does affect the investment decision if the quantity risk is smaller (see Exhibit 39.9, Example 2). As shown in the exhibit, the variation in price becomes the prevailing risk that drives the decision to start production in the case of lower variation in potential mine size outcomes. For example, if the price turns out to be high, production is started whether the mine size ends up at the higher or lower end of its range. As a result, the DTA approach now only provides an approximation (\$18.20) of the correct ROV value (\$19.50).¹⁶ Similarly, for some real-world investments, the nondiversifiable risks outweigh any technological, regulatory, or other diversifiable risks. For example, decisions to invest in the expansion of a power plant are typically driven by the difference between fuel and power prices and by overall demand for power.

¹⁴ Note that this value is higher than in the original example, because we now develop the mine only if the reserves are large; we have introduced additional flexibility.

¹⁵ If the probability distribution of the commodity price was continuous rather than discrete as in this example, there would always be some price outcome overturning the production decision. But the point remains that if the probability of reaching such price levels is small, the difference between the ROV and DTA outcome would be small, too.

¹⁶ For the ROV result, the present value of the large-mine and high-price scenario would now be $(0.45 \times 0.33 \times \$120)/(1 + 5\%) = \$17.1$. Adding this to the present value for the small-mine and high-price scenario, or $(0.45 \times 0.67 \times \$7.5)/(1 + 5\%) = \$2.3$, leads to total ROV of \$19.5 when rounded.

Data Availability: Traded vs. Untraded Assets The results of any contingent valuation critically depend on well-grounded estimates for the value and the variance of cash flows from the underlying asset.

If the estimate for the *underlying asset value* is inaccurate, the flexibility value also will be inaccurate. Returning to our first example, if we estimate incorrectly the future cash flows generated by a highly effective drug, the value of the option to defer will be inaccurate. In practice, you would have to estimate the value with a full-fledged DCF model projecting sales growth, operating margins, capital turnovers, and so on. All ROV (and DTA) approaches build on this valuation of the underlying asset.

A similar argument holds for estimates of the variance of the underlying asset's cash flows (called *volatility* in the option-pricing literature). Volatility can have a great impact on value, because real options typically have long lifetimes and are often at-the-money or close to it,¹⁷ meaning the decision of whether to undertake the project is a close call.¹⁸ Still, for many managers and practitioners, volatility remains an abstract concept: how do you reasonably estimate the range of cash flow outcomes from the sale of a product that has yet to be released?¹⁹

Sometimes the underlying asset value and variance can be derived from traded assets. Examples include options to shut down gas-fueled power generation, abandon a copper mine, or defer production of an oil field. In such cases, because you can estimate the key inputs with reasonable accuracy, ROV should be more accurate than DTA. When estimates for the underlying asset valuation and variance (volatility) cannot be derived from traded assets and are largely judgmental, a DTA approach is more appropriate. It is more straightforward and transparent to decision makers than the ROV approach. Transparency is especially important when critical valuation assumptions require the decision maker's judgment. DTA captures the essence of flexibility value, and the theoretical advantage of ROV is less important if required inputs are unavailable.

FOUR STEPS TO VALUING FLEXIBILITY

To value flexibility, use the four-step process illustrated in Exhibit 39.10. In step 1, conduct a valuation of the investment project without flexibility, using a traditional discounted-cash-flow model. In step 2, expand the DCF model into an event tree, mapping how the value of the project evolves over time, using (unadjusted) probabilities and the weighted average cost of capital. At this stage, the model does not include flexibility, so the present value of the

¹⁷ It follows from option-pricing theory that the sensitivity of option value to changes in variance (referred to as *vega*) increases as the option's lifetime increases and as the option is closer to the money. An option is at-the-money if its exercise price equals the value of the underlying asset.

¹⁸ If the investment decision were a clear go or no-go, there would be little value in flexibility in the first place, and no need to consider the option value.

¹⁹ The range needs to include the associated probabilities to provide a variance estimate.

EXHIBIT 39.10 Four-Step Process for Valuing Flexibility

	Estimate NPV without flexibility	Model uncertainty in event tree	Model flexibility in decision tree	Estimate contingent NPV
Objectives	Compute base-case present value without flexibility	Understand how present value develops with respect to changing uncertainty	Analyze event tree to identify and incorporate managerial flexibility to respond to new information	Value total project using DTA or ROV approach
Comments	Standard NPV approach is used for valuation of underlying asset.	No flexibility modeled; valuation following event tree should equal standard NPV	Flexibility is incorporated into event tree, transforming it into decision tree	Under high uncertainty and managerial flexibility, contingent NPV will be significantly higher than standard NPV

project, based on discounting the cash flows in the event tree, should still equal the standard DCF value from the first step.

In step 3, turn the event tree into a decision tree by identifying the types of managerial flexibility that are available. Build the flexibility into the nodes of the tree. Multiple sources of flexibility are possible at a single decision node, such as the option to abandon or expand, but it is important to have clear priorities among them. Be careful in establishing the sequence of decisions regarding flexibility, especially when the decision tree has compound options.

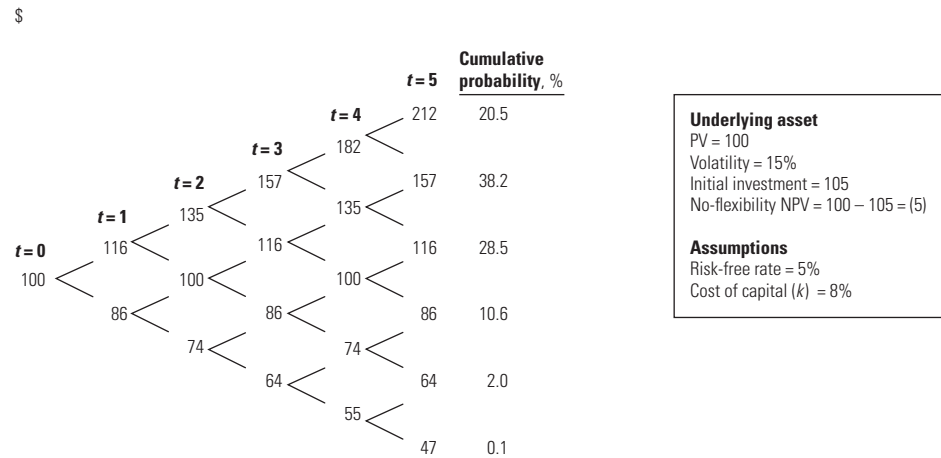
Finally, step 4 entails recognizing how the exercise of flexibility alters the project's risk characteristics. If the prevailing risk affecting the contingent cash flows is fully diversifiable, you need no special modeling; you can use DTA, discounting investment cash flows at the risk-free rate and the underlying project's cash flows at the weighted average cost of capital, as in the pharmaceutical example in the upcoming section on ROV and DTA. If the prevailing risk is nondiversifiable and priced in the market, the appropriate risk-adjusted discount rate for the project's cash flows is no longer the weighted average cost of capital used in step 1. In that case, apply an ROV approach for the project with flexibility, using risk-neutral valuation or a replicating portfolio.

Real-Option Valuation: A Numerical Example

Using the four-step process, we illustrate the ROV approach with a straightforward binomial lattice for valuing flexibility that is assumed to be driven by nondiversifiable risk. The results are identical to alternative option-pricing models that use more complicated mathematics such as stochastic calculus or Monte Carlo simulation.

Step 1: Estimate Net Present Value without Flexibility Assume that an investment in a project to build a factory generates cash flows whose present value (PV) equals \$100, and its expected rate of return and cost of capital (k) equal

EXHIBIT 39.11 **Event Tree: Factory without Flexibility**



Note: t = time, in years

8 percent. The risk-free rate is 5 percent per year, and the cash outflow necessary to undertake the project, if we invest in it immediately, is \$105. Thus, the standard NPV is –\$5, equal to the expected present value of \$100 less the investment of \$105, and we would not undertake the project if we had to commit today.

Step 2: Model Uncertainty Using Event Tree The lattice that models the potential values of the underlying risky asset is called an event tree. It contains no decision nodes and simply models the evolution of the underlying asset. Exhibit 39.11 illustrates potential values the factory might take for each of next five years, assuming a volatility of 15 percent per year.²⁰ Defining T as the number of years per upward movement and σ as the annualized volatility of the underlying factory value, determine the up-and-down movements by using the following formulas:²¹

$$\text{Up Movement} = u = e^{\sigma\sqrt{T}}$$

$$\text{Down Movement} = d = \frac{1}{u}$$

Substitute numerical values into these formulas:

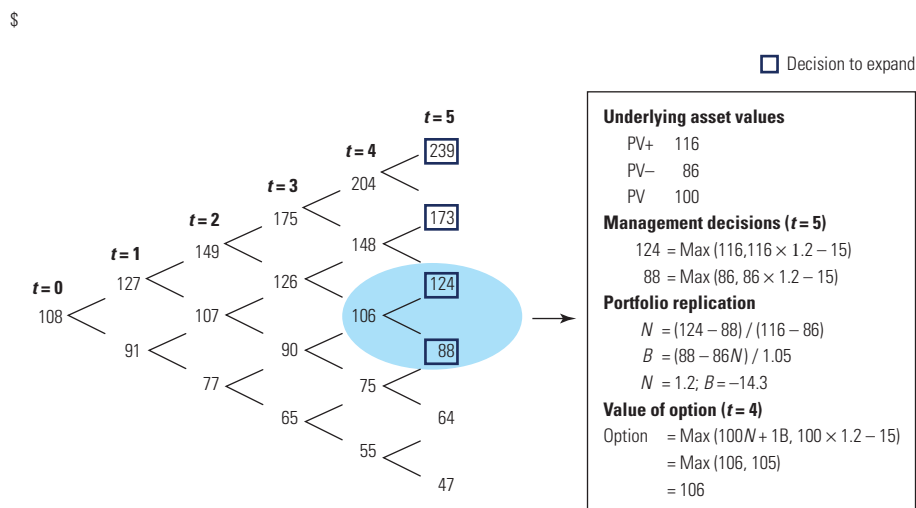
$$u = e^{0.15\sqrt{1}} = 1.1618$$

$$d = \frac{1}{1.1618} = 0.8607$$

²⁰ The standard deviation of the rate of change of the factory value.

²¹ J. Cox, M. Rubinstein, and S. Ross, "Option Pricing: A Simplified Approach," *Journal of Financial Economics* 7, no. 3 (1979): 229–263. As T becomes smaller, the binomial lattice results converge to the true value of the option. In this example, we have chosen $T = 1$ for ease of illustration.

EXHIBIT 39.12 Decision Tree: Option to Expand Factory



Note: t = time, in years
PV = present value
 N = number of replicating securities
 B = number of risk-free bonds
Incremental investment: \$15
Incremental payoff: 20%

Based on traditional DCF using an 8 percent cost of capital, the probability of an up movement is 72.82 percent, and the probability of a down movement is 27.18 percent.²² As can be verified, the present value of any branch in the event tree equals the expected payout discounted at the 8 percent cost of capital. For example, take the uppermost branch in the fifth time period. Its present value is:

$$PV_{t=4} = \frac{E(PV_{t=5})}{(1+k)} = \frac{0.7282(\$211.7) + 0.2718(\$156.8)}{1.08} = \$182.2$$

A similar calculation will produce any of the values in the event tree, resulting in a PV of the project of \$100 at $t = 0$. That present value equals the result in step 1, so we know the tree is correct.

Step 3: Model Flexibility Using a Decision Tree When you add decision points to an event tree, it becomes a decision tree. Suppose the factory can be expanded for an additional \$15. The expansion increases the factory's value at that node by 20 percent. The option can be exercised at any time during the next five years—but only once.

Exhibit 39.12 shows the resulting decision tree. To find the payouts at a given point on the tree, start with the final branches. Consider the uppermost

²² See the previous note for the derivation of the formula for estimating the upward probability:

$$\frac{(1+k)^T - d}{u - d} = \frac{(1+8\%)^5 - 0.8607}{1.1618 - 0.8607} = 0.7282$$

branch in period 5. On the upward limb, the payout absent expansion would be \$211.70, as Exhibit 39.11 shows. But with expansion, it is $1.20 \times \$211.70 - \$15 = \$239.00$. Since the value with expansion is higher, we would decide to expand. On the lower limb of that same node, the payout with expansion is $1.20 \times \$156.80 - \$15 = \$173.20$, versus \$156.80 without expansion, so again we would expand. In this way, complete the payoff estimates for all final branches.

Step 4: Estimate Contingent Net Present Value To determine the value of the project with the flexibility to expand, work backward through the decision tree, using the replicating-portfolio method at each node. For the node highlighted in Exhibit 39.12, you can replicate the payoffs from the option to expand in $t = 5$, using a portfolio of N units of the underlying project and B units of \$1 risk-free bonds:²³

$$\$116.2N + \$1.05B = \$124.4$$

$$\$86.1N + \$1.05B = \$88.3$$

Solving the equations, we find that $N = 1.2$, and $B = -14.3$. Therefore, a replicating portfolio consists of 1.2 units of the project without flexibility (at that node, valued at \$100 in the event tree of Exhibit 39.11), plus a short position of 14.3 bonds worth \$1. As shown in the calculations in Exhibit 39.12, the value of the option in the node at $t = 4$ is then:

$$PV = \$100N + \$1B = \$105.7$$

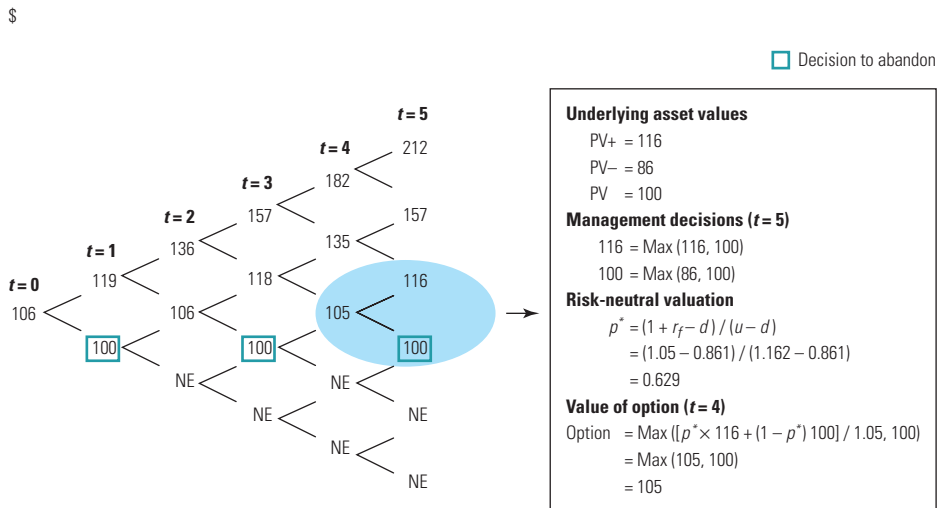
Work backward from right to left, node by node, to obtain a present value of \$108.40 for a project that has an option to expand. As a result, the net present value of the project increases from -\$5.00 to \$3.40, so the option itself is worth \$8.40. Note that the analysis also provides the value-maximizing decision strategy: management should expand the factory only after five years and only if the factory is worth \$75 or more as indicated by the boxed nodes in Exhibit 39.12.²⁴

If, instead, management had the option to abandon the factory at any node for a fixed liquidation value of \$100, the valuation would be as shown in Exhibit 39.13. Determine the contingent payoffs for final branches. Then work again from right to left through the decision tree. For the highlighted node at $t = 4$, the value of the underlying factory is \$116.20 in the upward branch and \$86.10 in the downward branch (see in the event tree of Exhibit 39.11). Given the ability to do so, the company would abandon the project for \$100

²³ If the project itself is not traded but a traded twin security exists, we can construct the portfolio in a similar way with units of the twin security and risk-free bonds.

²⁴ This is analogous to a call option on a stock that does not pay dividends: it is never exercised prematurely. For example, in the node highlighted in Exhibit 39.12, the value in year 4 of deferring the expansion of the factory to year 5 is \$105.70, as calculated in the preceding equation. The value of expanding in year 4 is $100 \times 1.20 - \$15 = \105 . It is therefore optimal to defer expansion, as is the case for all nodes before year 5.

EXHIBIT 39.13 Decision Tree: Option to Abandon Factory



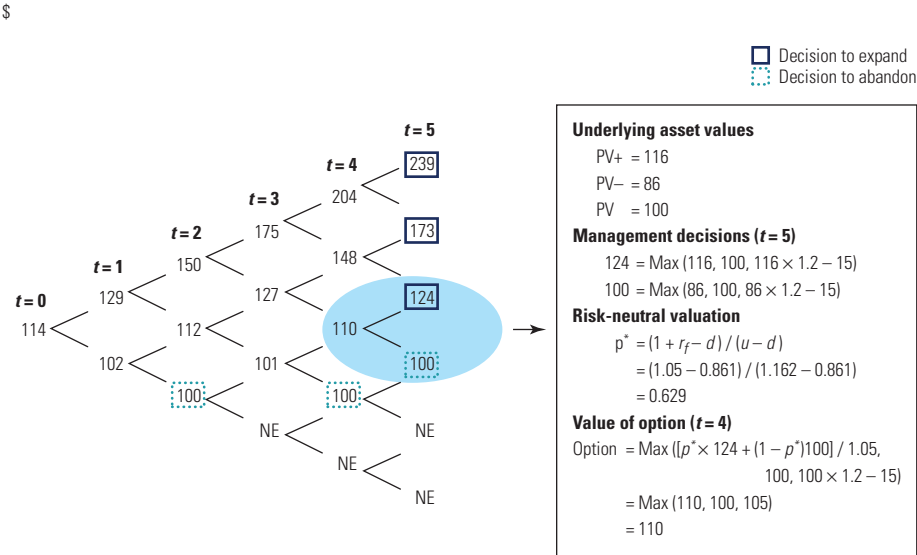
Note: t = time, in years
NE = nonexisting state
PV = present value
 p^* = binomial (risk-neutral) probability
 r_f = risk-free rate
 d = downward movement of value
 u = upward movement of value
Liquidation value: \$100

in the downward branch, so the payoffs in the decision tree are \$116.20 in the upward branch and \$100 in the downward branch. Using risk-neutral valuation this time, the abandonment option can be valued in the node at $t = 4$ at \$104.90, as shown in Exhibit 39.13 (the same result a replicating portfolio would have generated). Working backward through time, the value for a factory with the ability to abandon is \$106.40, so that the abandonment option is worth \$6.40. Now the value-maximizing decision strategy is to abandon the factory immediately in any year in which its value drops below \$100.

Multiple sources of flexibility can be combined within a single decision tree, as illustrated in Exhibit 39.14, using risk-neutral valuation. The value of the project, including the options to abandon and expand, would be \$113.50 rather than \$100, its stand-alone value without flexibility. With these options, the correct decision would be to accept the project. Note that the value of the combined expansion-abandonment flexibility, \$13.50, is less than the sum of the individual flexibility values (\$8.40 + \$6.40 = \$14.80) but greater than either of them individually. The values of both options are not additive, because they interact in complex ways (for example, you cannot expand the factory once you have abandoned it). As indicated in Exhibit 39.14, the best decision strategy is to abandon the factory whenever its value²⁵ drops below \$100 and to expand only in year 5 if its value exceeds \$75.

²⁵ Note that this is the value of the factory including the option to expand. Therefore, abandonment occurs only in more unfavorable states of the world than in Exhibit 39.13.

EXHIBIT 39.14 **Decision Tree: Option to Expand or Abandon Factory**



Note: t = time, in years
NE = nonexistent state
PV = present value
 p^* = binomial (risk-neutral) probability
 r_f = risk-free rate
 d = downward movement of value
 u = upward movement of value
Liquidation value: \$100
Incremental investment: \$15
Incremental payoff: 20%

REAL-OPTION VALUATION AND DECISION TREE ANALYSIS: A NUMERICAL EXAMPLE

Our next example applies both the DTA and the ROV approaches in the valuation of a research and development project. Assume a company needs to decide whether to develop a new pharmaceutical drug. In our simplified example,²⁶ the first step in development is a research phase of three years, in which the most promising chemical compounds are selected. The probability of success in the research phase is estimated at 15 percent. This is followed by a three-year testing phase, during which the compounds are tested in laboratory and clinical settings. The chance of successfully completing the testing phase is 40 percent. If there are successful results, the drug can be released in the market. On failure in any phase, the company terminates development, and the product dies worthless.

²⁶ Pharmaceutical R&D is much more complex and consists of more phases than shown in this example. For a more extensive example of valuing flexibility in pharmaceutical research and development, see Kellogg and Charnes, "Real-Options Valuation for a Biotechnology Company."

DTA Approach: Technological Risk

The DTA approach presented next follows the four steps for the valuation of flexibility as described in the previous section. In the DTA valuation of the research and development project, we consider only the prevailing technological risk relating to the research and testing outcomes. The commercial risk concerning the future profitability of the drug and the technological risk are taken into account jointly in the ROV approach discussed in the next section.

Step 1: Estimate Present Value without Flexibility If the development process succeeds, the drug will deliver substantial value in six years' time. Margins in the pharmaceutical industry are high because patents protect drugs against competition. A successful drug is expected to generate annual sales of \$2,925 million and 45 percent earnings before interest, taxes, depreciation, and amortization (EBITDA) margin on sales until its patent expires, ten years after its market launch. (Because prices decline drastically after a patent expires, we do not count cash flows beyond that time.) Assuming a 30 percent tax rate and a 7 percent cost of capital, a marketable drug's present value at the launch date would therefore be \$6,475 million. Unfortunately, the odds of successful development are small. The cumulative probability of success over the research and testing phase is only 6 percent (0.15 for research $\times$ 0.40 for testing). In addition, the investments needed to develop, test, and market a drug are high: \$100 million in the research phase, \$250 million in the testing phase, and \$150 million in marketing.

If we had to commit to all three investments today, we should not proceed, because the NPV would be negative:

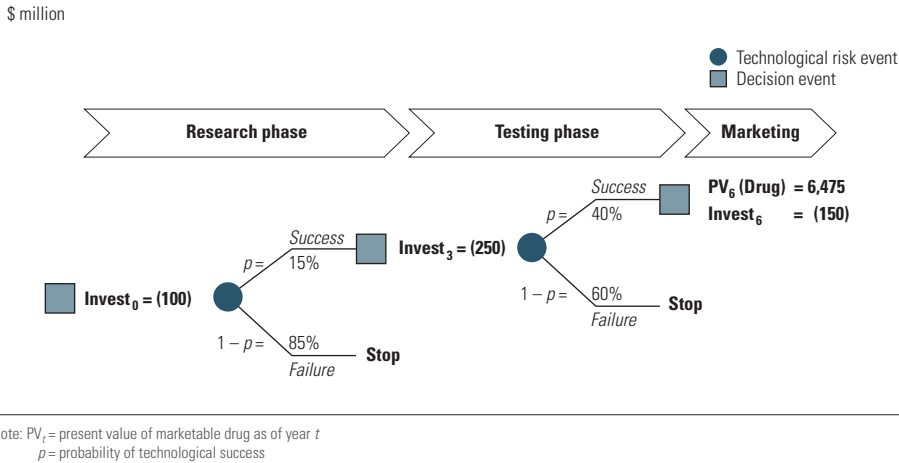
$$\begin{aligned}\text{Standard NPV}_0 &= \text{PV}_0(\text{Expected Cash Flows}) - \text{PV}_0(\text{Investments}) \\ &= 0.06 \left[\frac{\$6,475}{(1.07)^6} \right] - \$100 - \frac{\$250}{(1.05)^3} - \frac{\$150}{(1.05)^6} \\ &= -\$169\end{aligned}$$

However, if we take into account management's ability to abandon the project before completion, the value is significantly higher.

Step 2: Model Uncertainty Using an Event Tree For this development project, you can model the prevailing technological risk using a straightforward event tree (see Exhibit 39.15). The expected value of a marketable drug after successful development is shown at its DCF value of \$6,475 million as of $t = 6$.

Step 3: Model Flexibility Using a Decision Tree Next, include decision flexibility in the tree, working from right to left. At the end of the testing phase, we have the option to invest \$150 million in marketing to launch the product.

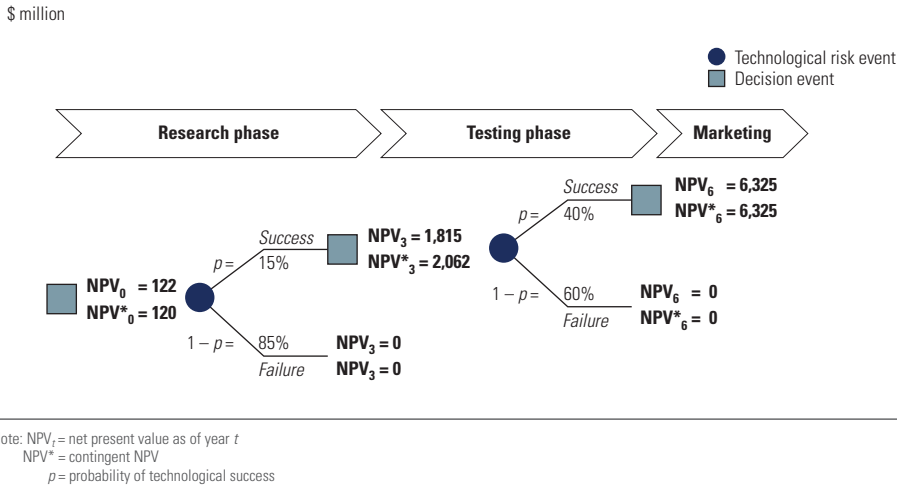
EXHIBIT 39.15 Event Tree: R&D Option with Technological Risk



We should invest only if testing has produced a marketable product. At the end of the research phase, we have the option to proceed with the testing phase. We proceed to testing only if the payoffs justify the incremental investment of \$250 million.

Step 4: Estimate Value of Flexibility Because the technological risk is fully diversifiable, apply a straightforward DTA approach for the valuation of flexibility. Again, work from right to left in the tree (see Exhibit 39.16). After six years, at the end of the testing phase, we proceed with launching the product only if there is a marketable product. The value in millions at this point in time is therefore $NPV_6 = \text{Max}[(\$6,475 - \$150), 0] = \$6,325$. At the end of the research

EXHIBIT 39.16 Decision Tree: R&D Option with Technological Risk



phase, we continue to the testing phase if the future payoffs outweigh the required investments. The value of the project at this point, after three years is:

$$NPV_3(\text{Option}) = \text{Max} [PV_3(\text{Testing}) - \text{Inv}_3(\text{Testing}), 0]$$

In this equation, $PV_3(\text{Testing})$ equals the probability-weighted future payoffs discounted by three years at the cost of capital of 7 percent:

$$PV_3(\text{Testing}) = 0.40 \left[\frac{\$6,475 - \$150}{(1.07)^3} \right] + 0.60(0) = \$2,065$$

With $\text{Inv}_3(\text{Testing})$ equal to the \$250 million investment requirement for the testing phase, the project value prior to the testing phase amounts to:

$$NPV_3(\text{Option}) = \text{Max} [(\$2,065 - \$250), 0] = \$1,815$$

Working further from right to left in the tree, we find the contingent NPV for the entire project prior to the research phase:

$$\begin{aligned} NPV_0(\text{Option}) &= \text{Max} [PV_0(\text{Research}) - \text{Inv}_0(\text{Research}), 0] \\ &= \text{Max} \left[0.15 \left(\frac{\$1,815}{(1.07)^3} \right) + 0.85(0) - \$100, 0 \right] \\ &= \$122 \end{aligned}$$

This value including flexibility is significantly higher than the standard NPV of $-\$169$ million. Note that we discounted all contingent payoffs at the underlying asset's cost of capital, so the \$122 million only approximates the true contingent value. But the result is close, as we show in the calculations immediately following, and this approach is straightforward to apply and easy to explain.

The true contingent value turns out to be \$120 million and follows from a refined DTA approach that separately discounts the asset cash flows at the cost of capital of 7 percent and the investment cash flows at the risk-free rate of 5 percent.²⁷ The value of proceeding with testing now becomes:²⁸

$$PV_3^*(\text{Testing}) = 0.40 \left(\frac{\$6,475}{(1.07)^3} - \frac{\$150}{(1.05)^3} \right) + 0.60(0) = \$2,114 - \$52 = \$2,062$$

The value of the option to proceed with the testing phase is then:

$$NPV_3^*(\text{Option}) = \text{Max} [(\$2,114 - \$52) - \$250, 0] = \$2,114 - \$302 = \$1,812$$

²⁷ See the example in Exhibit 39.7. The assumption to discount investment outlays at the risk-free rate is also implicitly made in ROV approaches.

²⁸ In prior editions of this book, we adopted an alternative but equivalent decision tree where all values of asset and investment cash flows were discounted to $t = 0$ before deriving the contingent value by working from right to left in the tree. The contingent NPV results are identical.

Working from right to left but now separately discounting asset and investment cash flows in each step, we obtain the contingent NPV* per $t = 0$:

$$\begin{aligned}\text{NPV}_0^*(\text{Option}) &= \text{Max} [\text{PV}_0^*(\text{Research}) - \text{Inv}_0(\text{Research}), 0] \\ &= \text{Max} \left[0.15 \left(\frac{\$2,114}{(1.07)^3} - \frac{\$302}{(1.05)^3} \right) + 0.85(0) - \$100, 0 \right] \\ &= \$120\end{aligned}$$

To illustrate, we obtain the same value of \$120 million with yet another approach: the ROV method. In this approach, project the future value of the underlying asset under “risk-neutral” return assumptions, and then discount all contingent payoffs at the risk-free rate. The risk-neutral future value of a successfully developed drug is its value as of today, compounded at the risk-free rate for six years:

$$\text{PV}_6^{**}(\text{Drug}) = \$4,314(1.05)^6 = \$5,781$$

This means that the risk-neutral value of proceeding with testing is:

$$\text{PV}_3^{**}(\text{Testing}) = 0.40 \left(\frac{\$5,781 - \$150}{(1.05)^3} \right) + 0.60(0) = \$1,946$$

The risk-neutral value of the option to proceed with testing as of $t = 3$ is:

$$\text{NPV}_3^{**}(\text{Option}) = \text{Max}[\$1,946 - \$250, 0] = \$1,696$$

Working from right to left while discounting all cash flows at the risk-free rate gives us the contingent NPV at $t = 0$, which is \$120 million:

$$\begin{aligned}\text{NPV}_0^{**}(\text{Option}) &= \text{Max}[\text{PV}_0^{**}(\text{Research}) - \text{Inv}_0(\text{Research}), 0] \\ &= \text{Max} \left[0.15 \left(\frac{\$1,696}{(1.05)^3} \right) + 0.85(0) - \$100, 0 \right] \\ &= \$120\end{aligned}$$

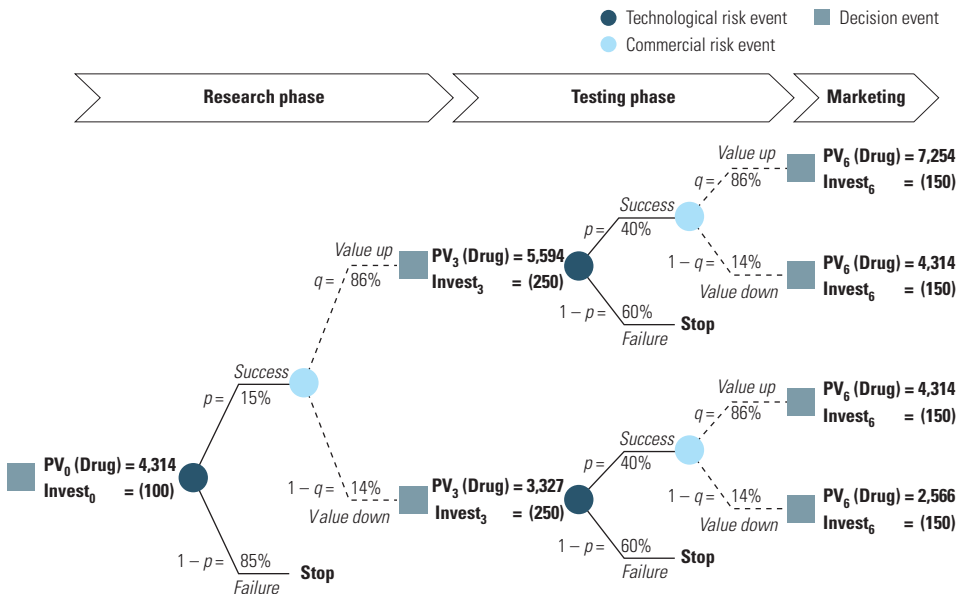
ROV Approach: Technological and Commercial Risk

Our analysis thus far did not include the other source of uncertainty in the development project: the commercial risk concerning the future cash flow potential of the successfully developed and marketed drug. ROV is necessary to handle both technological and commercial risk.

Step 1: Estimate Present Value without Flexibility The first step, estimating present value without flexibility, is identical for the DTA and ROV approaches.

EXHIBIT 39.17 Event Tree: R&D Option with Technological and Commercial Risk

\$ million



Note: $PV_t(\text{Drug})$ = present value of marketable drug as of year t
 Invest_t = investment as of year t
 p = probability of technological success
 q = probability of drug value increase

Step 2: Model Uncertainty Using an Event Tree Both risks can be modeled in a combined event tree (see Exhibit 39.17). For simplicity, we have chosen a one-step binomial lattice to describe the evolution of the drug value over each three-year period.²⁹ Assuming an annual volatility of 15 percent, we can derive the upward and downward movements, u and d , as follows:

$$u = e^{\sigma\sqrt{T}} = e^{0.15\sqrt{3}} = 1.30$$

$$d = \frac{1}{u} = \frac{1}{1.30} = 0.77$$

The probability of an upward movement is 86 percent, and the probability of a downward movement is 14 percent.³⁰ The value of a marketable drug

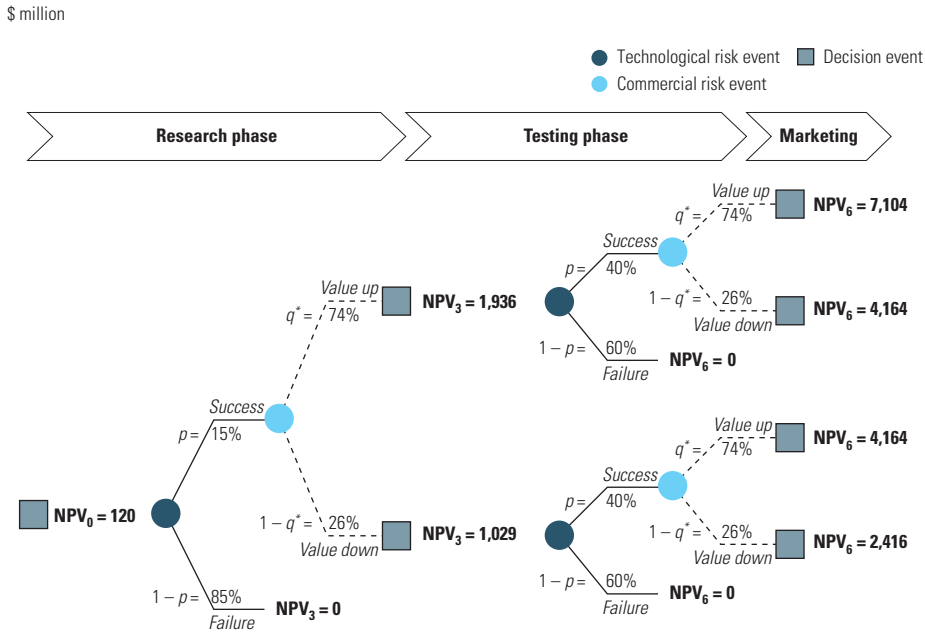
²⁹ With more nodes, the tree quickly becomes too complex to show in an exhibit, because it does not converge in the technological risk. We carried out the analysis with ten nodes and found that doing so did not affect the results for this particular example.

³⁰ The formula for estimating the upward probability is:

$$\frac{(1+k)^T - d}{u - d} = \frac{1.07^3 - 0.77}{1.30 - 0.77} = 0.86$$

where k is the expected return on the asset.

EXHIBIT 39.18 **Decision Tree: R&D Option with Technological and Commercial Risk**



Note: NPV_t = net present value of project as of year t
 q^* = binomial (risk-neutral) probability of an increase in marketable drug value
 p = probability of technological success

at the start of the research phase is \$4,314 million. At the end of the research phase, there are three possible outcomes: success combined with an increase in the value of a marketable drug to \$5,594 million, success combined with a decrease in the value of a marketable drug to \$3,327 million, and failure leading to a drug value of \$0. Following the same logic, there are six possible outcomes after the testing phase.

Step 3: Model Flexibility Using a Decision Tree The logic underlying the decision tree including commercial risk (see Exhibit 39.18) is the same as under the DTA approach. For example, the payoff at the end of the testing phase in the top branch equals $\text{Max}[(\$7,254 - \$150), 0] = \$7,104$. The primary difference is that the ROV version of the tree recognizes the ability to abandon development if the value of a marketable drug drops too much.

Step 4: Estimate Contingent NPV The commercial risk regarding the drug's future cash flows is not diversifiable,³¹ so you need to use an ROV approach to include it in your valuation. This example uses risk-neutral valuation.

³¹ Recall that we assumed the cost of capital for a marketed drug is 7 percent. Given our assumption for a risk-free rate of 5 percent, its beta must be different from zero.

Therefore, risk-adjust all probabilities of the upward and downward movements for the drug's value:

$$p^* = \frac{(1 + r_f)^T - d}{u - d} = \frac{1.05^3 - 0.77}{1.30 - 0.77} = 0.74$$

Having applied the risk-neutral probabilities, discount all contingent payoffs at the risk-free rate, working from right to left in the tree. Because the technological risk is fully diversifiable, there is no need to adjust the probabilities for success and failure in research or testing.

For example, from Exhibit 39.18, the value of the option at the end of the research phase showing a drop in the value of the drug is expressed as follows:

$$NPV_3(\text{Option}) = \text{Max}[PV_3(\text{Testing}) - \text{Inv}_3(\text{Testing}), 0]$$

In this equation, $PV_3(\text{Testing})$ represents the value of proceeding with testing at this node. It equals the value of the future payoffs weighted by risk-neutral probabilities and discounted at the risk-free rate:

$$PV_3(\text{Testing}) = \frac{0.40[0.74(\$4,164) + 0.26(\$2,416)] + 0.60(0)}{(1.05)^3} = \$1,279$$

$\text{Inv}_3(\text{Testing})$ equals \$250 million, so the value of the development project at this node is as follows:

$$NPV_3(\text{Option}) = \text{Max}[(\$1,279 - \$250), 0] = \$1,029$$

Solve for the other nodes in the same way. Working backward through the tree gives us an estimate of the contingent NPV: \$120 million, the same result as obtained in the DTA approach without commercial risk.

This is not surprising. A closer look at the decision tree reveals that uncertainty about the future value of the drug if it is marketable is not significant enough to influence any of the decisions in the development process. In this example, the commercial risk makes no difference, even if we assume volatility as high as 50 percent (an amount that exceeds the volatility of many high-tech stocks). As noted earlier, when nondiversifiable risk (the drug's commercial risk as measured by its beta) does not influence investment decisions, the DTA and ROV results are equivalent.

Moreover, in real situations, the prevailing uncertainty in drug development is whether the drug proves to be an effective disease treatment without serious side effects. The commercial risk is far less relevant, because a truly effective drug almost always generates attractive margins. The example illustrates how in such cases it is more practical to focus on the technological risk entirely, using a DTA approach. Explicitly modeling the nondiversifiable (e.g., commercial) risk requires an ROV approach that is more complex and may not even affect the valuation results.

In general, when faced with multiple sources of underlying risk, carefully assess whether all of these possible risks are important or whether one prevails. Sometimes you can focus the valuation approach on just one or two sources of uncertainty and greatly simplify the analysis.

SUMMARY

Managerial flexibility lets executives defer or change investment decisions as a business or project develops. It can substantially alter the value of a business or project. Rigidly applying standard DCF analysis fails to account for the impact that exercising flexibility can have on present value.

Flexibility takes many forms, such as the option to defer, expand, contract, or abandon projects, or to switch them on and off. This chapter has illustrated only a few applications. Contingent NPV analysis, in the form of decision tree analysis (DTA) or real-option valuation (ROV) models, correctly captures flexibility's impact on value. The ROV approach is theoretically superior to DTA, but applying it is more complicated. So ROV is often limited to valuing flexibility in commodity-based industries where prices are measurable, making its application more straightforward. In most other cases, a careful DTA approach delivers results that are reasonably solid and can provide more valuable insights.

REVIEW QUESTIONS

1. Define contingent net present value (NPV). Outline and explain the differences between standard and contingent NPV.
2. Identify the value drivers embedded in a “real” option. What is each driver’s effect on the option’s value, and how might they interact?
3. Assume a company runs a plant for which the value one year from now is either \$1,000 if market growth is positive or \$250 if market growth is negative. The probability of positive market growth is 60 percent, and the probability of negative market growth is 40 percent. At any time, the company can choose to close the plant and collect the salvage value of \$285 if scrapped today or \$300 if scrapped in one year. The cost of capital for the plant is 10 percent, and the risk-free rate is 5 percent. Estimate the value of the plant using the standard NPV, decision tree analysis (DTA), and real-option valuation (ROV) valuation models. Explain the differences in results.
4. Under what circumstances should a manager apply a standard NPV approach, a DTA approach, or an ROV approach to valuation?
5. It is often argued that the two most important real options available to a manager evaluating investment decisions are the option to defer an investment decision and the option to abandon an investment decision. Explain the significance of these two options. What insights could the ROV or the DTA model provide into these decisions?
6. The option to defer an investment reduces risk for a company because it does not need to commit the full investment outlay until there is more certainty about the true value of the underlying asset. But the implied cost of capital for the project including flexibility is higher than for the project without flexibility (see, e.g., Exhibit 39.7). Explain why this is the case.
7. For estimating the value of an option on a traded stock, the expected return on the stock is irrelevant, as proven in option-pricing theory. For the valuation of an option on an asset that is not traded, such as in the numerical example introduced in Exhibit 39.11, the expected cash flow returns are required. Discuss how that is still consistent with option-pricing theory.
8. Consider the example of the valuation of the pharmaceutical R&D project described in the final section of the chapter. Under the assumptions stated, the DTA value is identical to the ROV value. Calculate what volatility (as modeled in terms of parameters u and d) would make the ROV value differ from the DTA value, and discuss what drives this difference.

